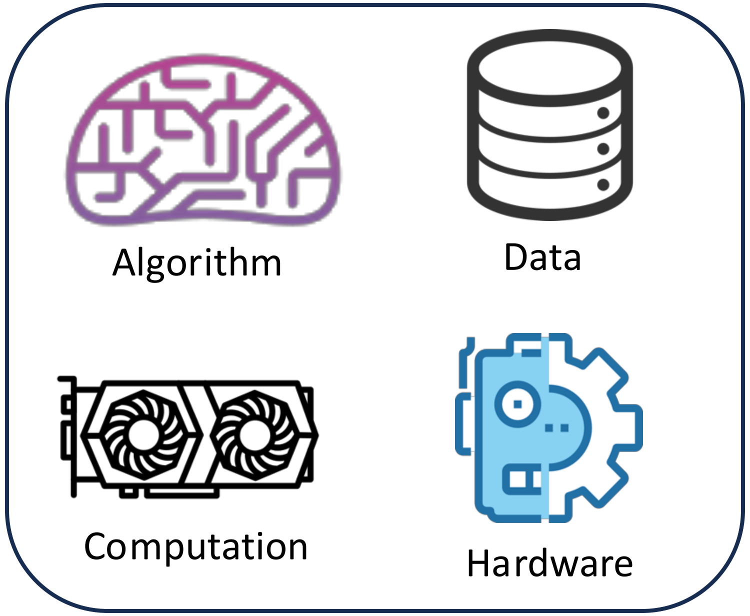
16-831: Introduction to Robot Learning, Fall 2024

Robot Learning: What, Why, and How?



Embodied intelligence in the physical world

Instructor: Guanya Shi

LeCAR
Learning and Control for Agile Robotics Lab

Carnegie Mellon University
The Robotics Institute

Recording Reminder

- This session will be audio recorded for educational use by other students in this course.

Welcome to the First Day!

Logistics

- There are ~50 students on the waitlist
  - I know!

- It is hard to increase the volume
  - We have projects (40% grade is from project, 20% pre + 20% report)
  - Interaction/discussion is important for learning

- We will offer the same class in Spring 2025 (I will teach it)
  - Please consider taking the spring one!
  - If you plan to drop, don’t wait until deadline

- Feel free to audit!

Logistics

- No Canvas. Only (1) Website, (2) Piazza, and (3) Gradescope
- Website: https://16-831.github.io/
- Piazza: https://piazza.com/cmu/fall2024/16831/resources
  - Sign up if you haven't been enrolled yet!
- Gradescope: https://www.gradescope.com/courses/812979
  - Sign up if you haven't been enrolled yet!

Logistics

- This course will be recorded but only for students in this course
- Some slides are from Deepak Pathak (F23), David Held (F22), and Katerina Fragkiadaki (10-703), and other courses
- Speak up and participate! You can interrupt me at anytime.

Take Care of Yourself!

- It is natural to struggle with stress and mental health challenges.
- All of us benefit from support during times of struggle.
- You are not alone.
- Asking for support sooner rather than later is often helpful.
- Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit their website at http://www.cmu.edu/counseling/.

- Feel free to use my OH (Tue 11:00-11:45 AM @ NSH 3209) to chat about life, career plan, research ideas, and more!
- I will stay 5-10mins after each class

Course Organizations

- All the details will be on the website
  - Office hour (OH)
  - Latest course schedule and due dates
  - Optional reading (please do read them!)
  - Late policy (5 late days, but ONLY for four HW assignments), collaboration policy, academic integrity, etc.

- Assessment:
  - 60% Four assignments (15% each)
  - 20% Project presentation (on Dec 03 and 05)
  - 20% Project final report

Course Structure

Latest schedule: https://16-831.github.io/fall24/lectures

- Topic group 1: Course introduction: What is robot learning and why?
- Topic group 2: Machine/Deep learning refresher
- Topic group 3: Imitation learning
- Topic group 4: Model-free reinforcement learning
- Topic group 5: Model-based reinforcement learning
- Topic group 6: Bandit, preference-based learning, exploration
- Topic group 7: Reinforcement learning from offline data (one guest lecture)
- Topic group 8: Specialized topics (one guest lecture)
- Project presentation

Course Project

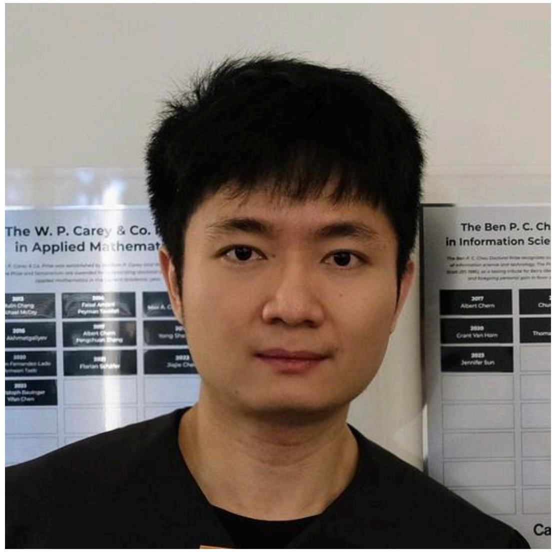
- Team size: 3-5 people (team size 1-2 is only allowed under exceptional circumstances).
- Two tracks:
  - Reproduce an existing robot learning paper and implement some improvement ideas on top of it (the ideas may or may not work)
  - Open-ended project
- Three components:
  - 1-2 page project proposal (due on Oct 10): It will carry NO grade, but is mandatory to submit for early feedback.
  - Project oral presentation (Dec 3 and 5)
  - 6-8 page project final report (due on Dec 13, in CoRL template)
- CMU SCS might be able to offer some AWS credits (<$150 per student). We will make a Google form soon to collect information.

Guanya Shi

- ❑ Ph.D. from Caltech (Control and Dynamical Systems)
- ❑ Assistant Professor at CMU RI from 2023 Sep
- ❑ Research area: learning and control for agile robotics
    - Theory and foundations
    - Algorithms
    - Real-world applications in robotics and autonomy
    - Robots: drone, car, quadruped, mobile manipulator, humanoid
- ❑ Email: guanyas@andrew.cmu.edu
- ❑ OH: Tue 11:00-11:45 AM @ NSH 3209

LeCAR

Learning and Control for Agile Robotics Lab



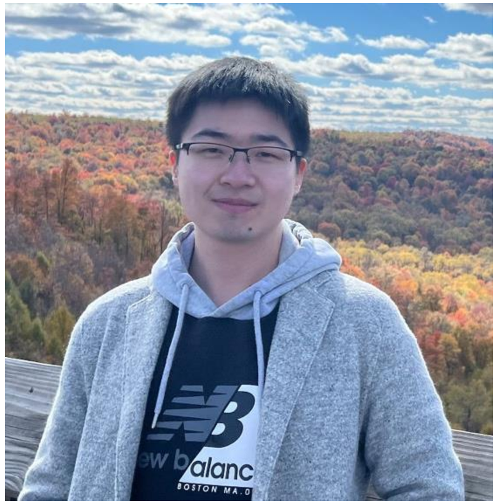
TA Team

Tairan He

Yilin Wu

Eliot Xing

Yufei Wang



Bowen Li

Tairan He

Hello!

- ❑ Second-year PhD student in RI
- ❑ Advised by Guanya Shi and Changliu Liu
- ❑ Work on learning-based control, legged robots, humanoid loco-manipulation
- ❑ Fun facts: The humanoid in the photo broke my laptop screen thanks to my not-so-great RL policy — so mastering this course could save your laptop too! 😄
- ❑ E-mail: tairanh@andrew.cmu.edu
- ❑ OH: Fridays 2:00 pm - 2:45 pm, NSH 3208

Yilin Wu

Hello!

- Second-year PhD student in RI
- Advised by Andrea Bajcsy (Intent Lab)
- Work on human robot interaction and learning-based manipulation.
- Fun facts: I am a huge fan of Roger Federer but I don't play tennis
- E-mail: yilinwu@andrew.cmu.edu
- OH: Tuesdays 2:00 pm - 2:45 pm, NSH 3104

Eliot Xing

Hello!

- Third-year PhD student in RI
- Advised by Jean Oh
- Work on reinforcement learning and deformable object manipulation
- E-mail: etaoxing@cmu.edu
- OH: Fridays 11:30am - 12:15pm

Yufei Wang

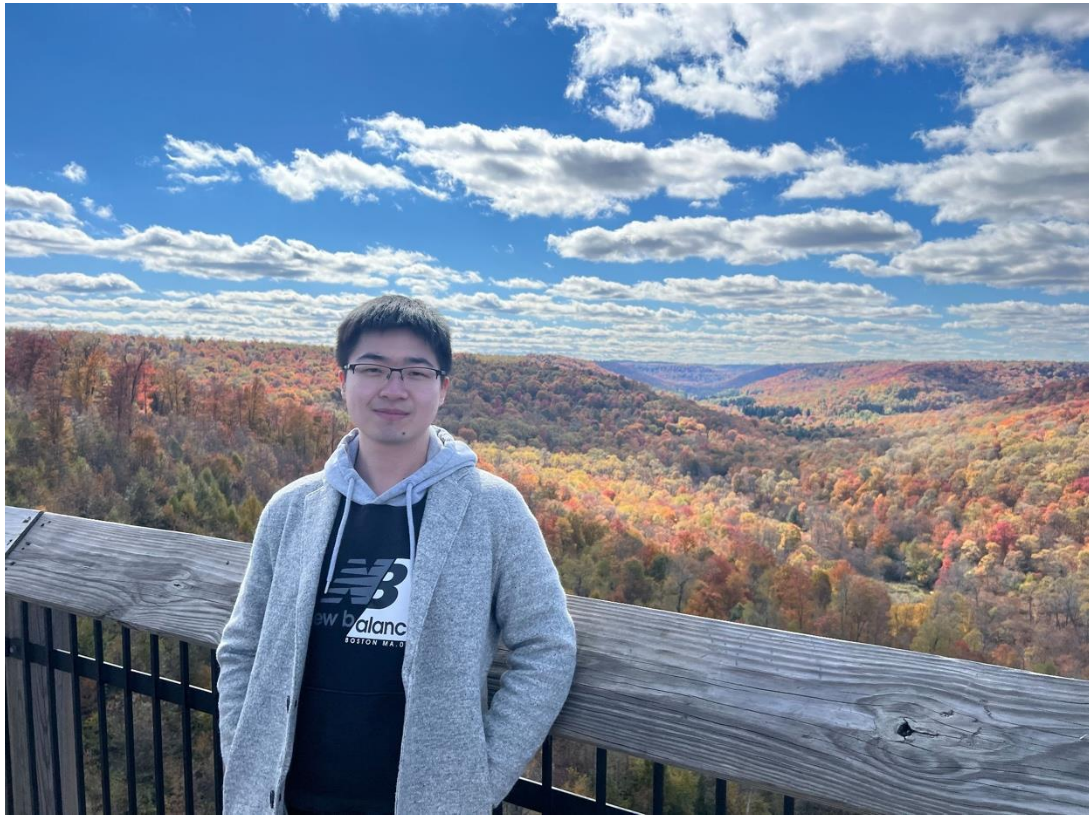
Hello!

- 4th-year PhD student in RI
- Advised by David Held and Zackory Erickson
- Work on robot learning for manipulation and assistive robots
- Fun facts: My dog is named Mars since he is a red-merle border collie and 4th in his litter (Mars is red and the 4th planet in the solar system)
- E-mail: yufeiw2@andrew.cmu.edu
- OH: Fridays 2:15 pm - 3 pm, NSH 4126

Bowen Li

Hello!

- Third-year PhD student in RI
- Advised by Sebastian Scherer (AirLab)
- Work on neuro-symbolic (abstract) planning/reasoning.
- Fun facts: The picture is taken in Kinzua Bridge, very beautiful at fall time
- E-mail: bowenli2@andrew.cmu.edu
- OH: Thursdays 11:15 am - 12:00 pm, NSH 1505



Quick Introductions

Please make your introduction <10s. Thanks!

- ❑ Mention if you are on the waitlist
- ❑ Name / advisor or group or major / what do you work on?
- ❑ Very important question: Is microwave a robot?

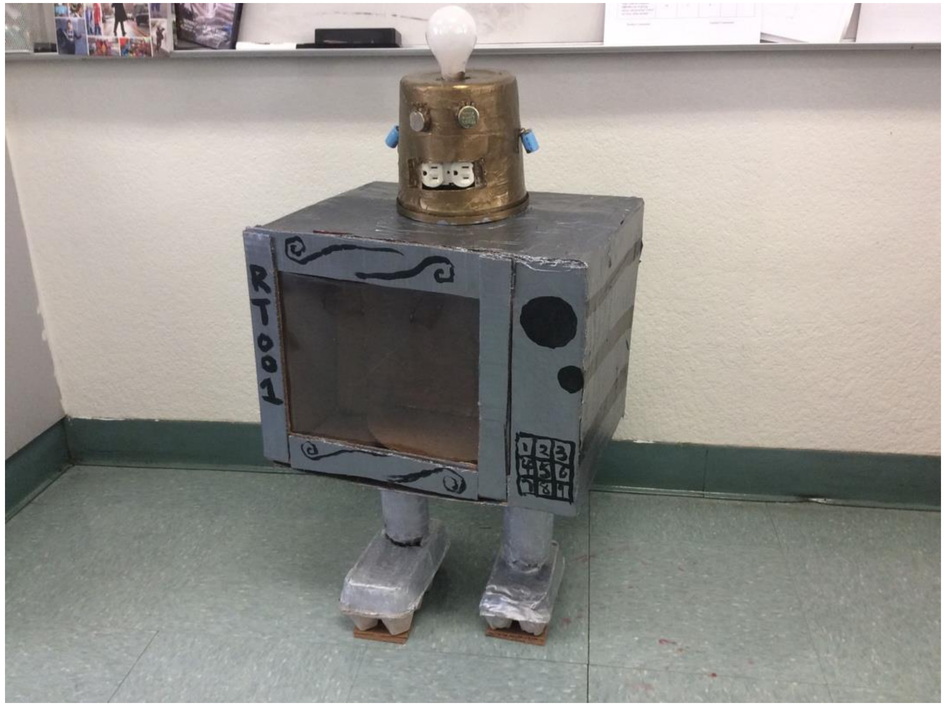


Photo credit: https://www.pinterest.com/pin/microwave-food-creator-robot-rt001-issue-famine--508977195364711362/

Lecture 1&2's Agenda

- ❑ What is robot learning and what is its uniqueness?
- ❑ What is the goal of robot learning?
- ❑ Where are we today? How far are we from the goal?
- ❑ Course structure and course objective
- ❑ Relation to other CMU courses
- ❑ Applications of sequential decision making in other domains

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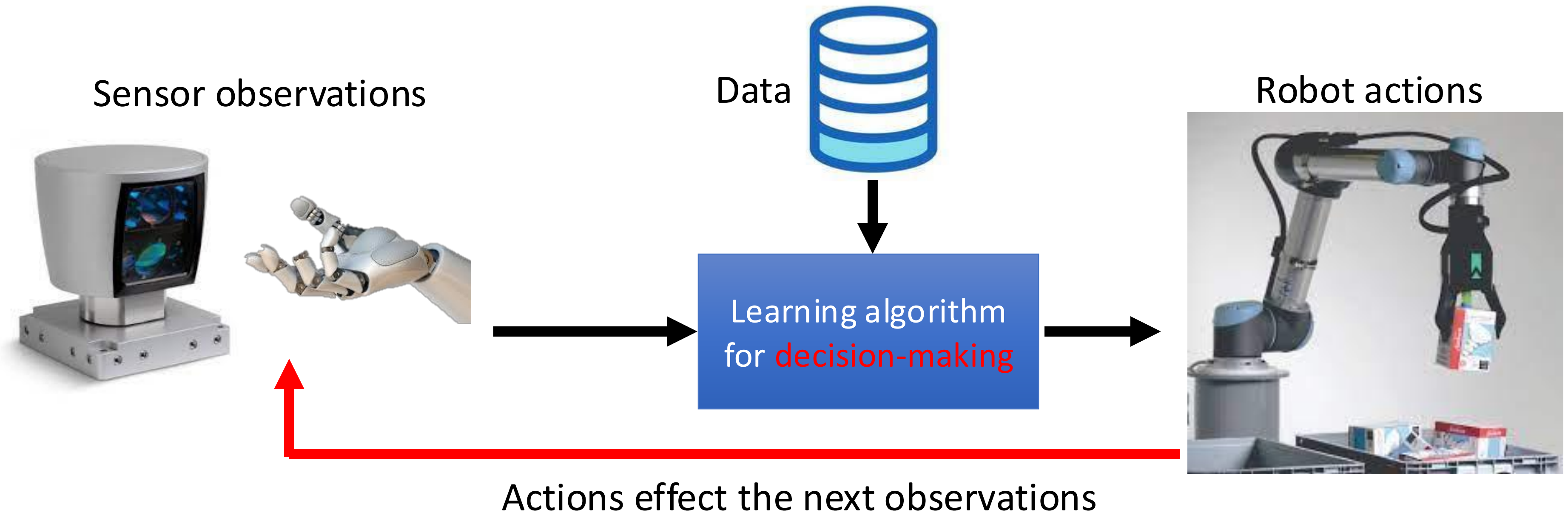
Let’s Start!

What is “robot learning” and what is this class about?

- ❑ **Learning** to make **sequential** decisions in the **physical world**



What is “Robot Learning”?

- **Learning** to make **sequential** decisions in the **physical world**

- **Learning**: Data-driven and improve from data
  - W/o learning & data: search & planning, classic control, …

- **Sequential**: The current action/decision influences the next state therefore the next action/decision
  - W/o sequential: bandit, standard supervised learning, …

- **Physical world**: The robot needs to interact with the physical world in the closed-loop
  - A.k.a. “embodied intelligence”
  - W/o physical world: RL for games, LLMs…

Uniqueness of "Robot Learning"

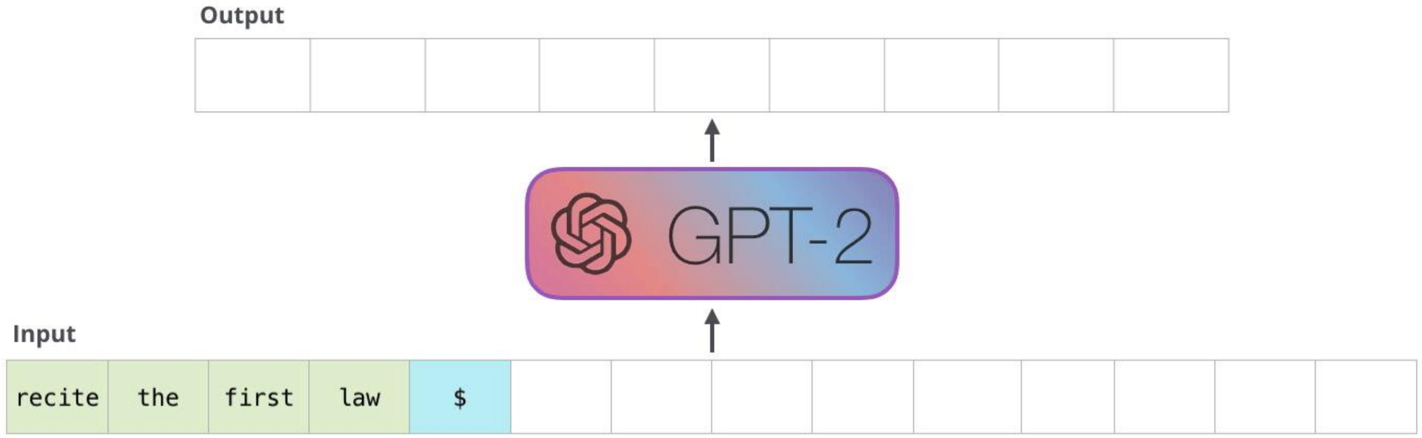
- **Learning**: Data-driven and improve from data
  - W/o learning & data: search & planning, optimization, classic control, …

Challenges (compared to other AI or machine learning domains):

- Where is the data from?
- How to use the data?

Why do GPTs (and other VLMs/LLMs) work? How much does it apply to robot learning?

- Architecture: transformer
- Data: web texts, books, wikis, …
- Loss: next token prediction
- Optimization: SGD
- Generation: autoregressive



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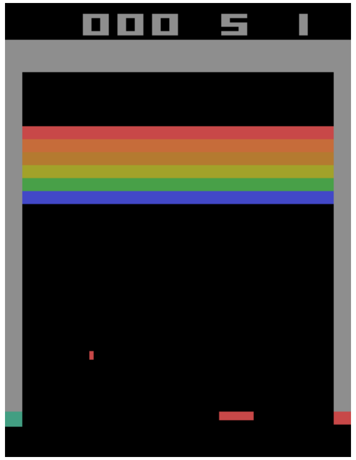
  - W/o physical world: RL for game, …

What are challenges (compared to DRL in games such as Alpha Go)?



DQN for Atari (DeepMind)

AlphaGo (DeepMind)

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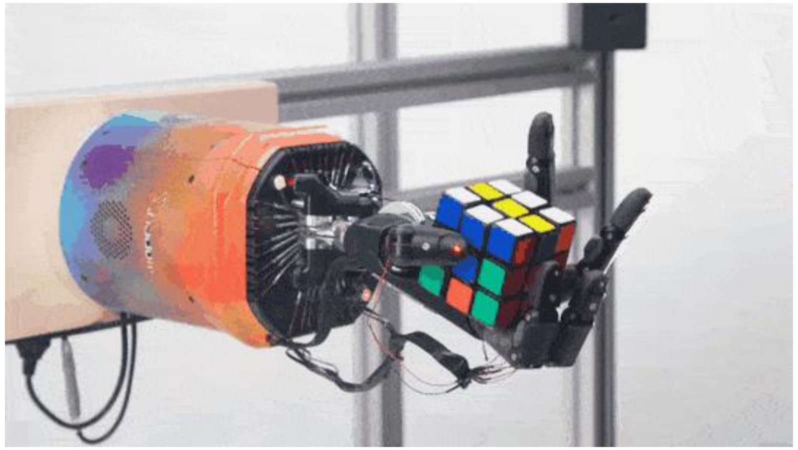

Manipulator (OpenAI)

Uniqueness of “Robot Learning”

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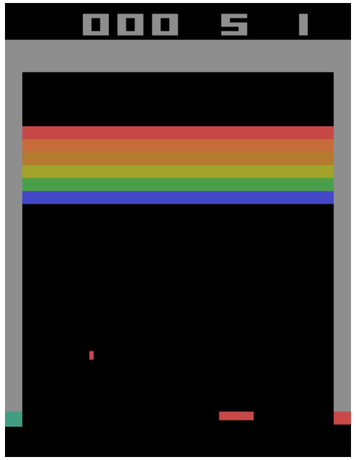

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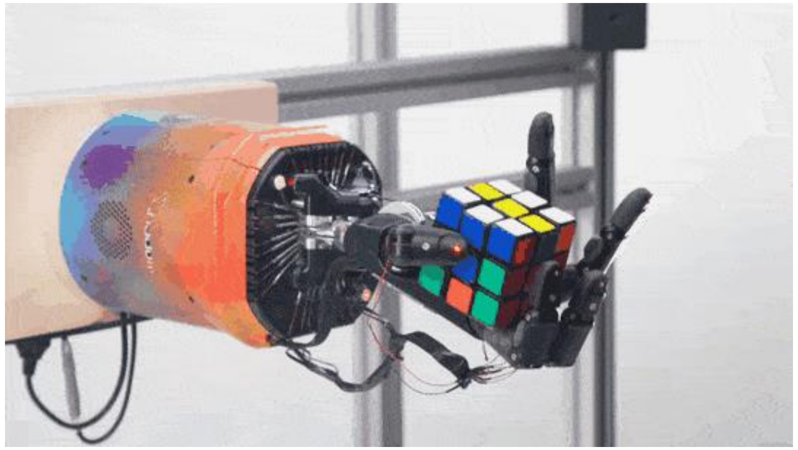

Manipulator (OpenAI)

- ❑ Known & static v.s. Unknown & dynamic
- ❑ One specific task v.s. Many tasks
- ❑ Clear optimization goals (i.e., reward) v.s. Unclear goals
- ❑ Offline learning is enough v.s. Require online adaptation
- ❑ Slow action v.s. Relatively fast real-time action (e.g., 50Hz)
- ❑ Allow failures v.s. Physics doesn’t forgive!
- ❑ Discrete digital world v.s. Continuous physical world

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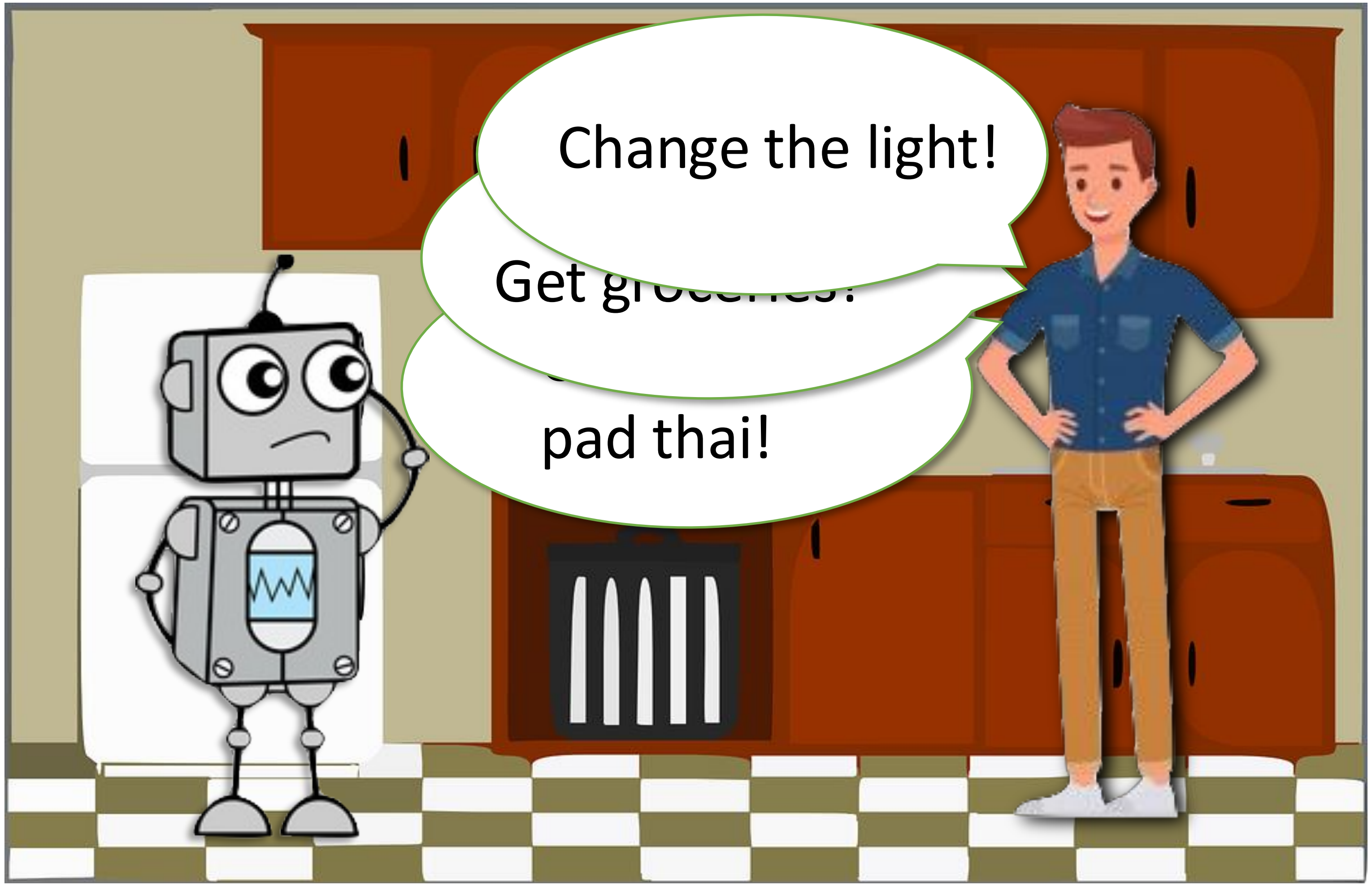
What is the Goal of “Robot Learning”?

- Ultimate goal: Build **general-purpose** **embodied** intelligence by learning to make sequential decisions in the physical world



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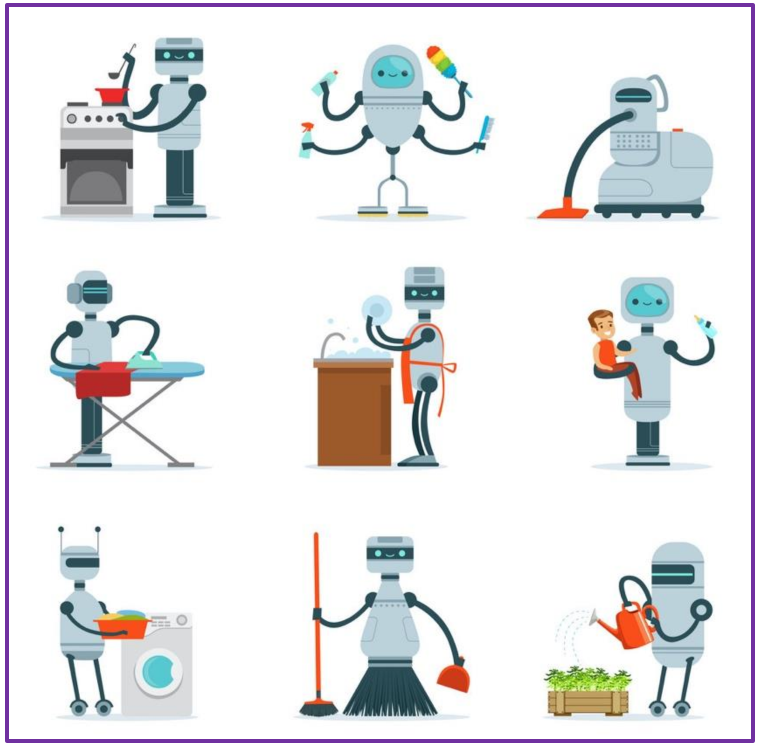
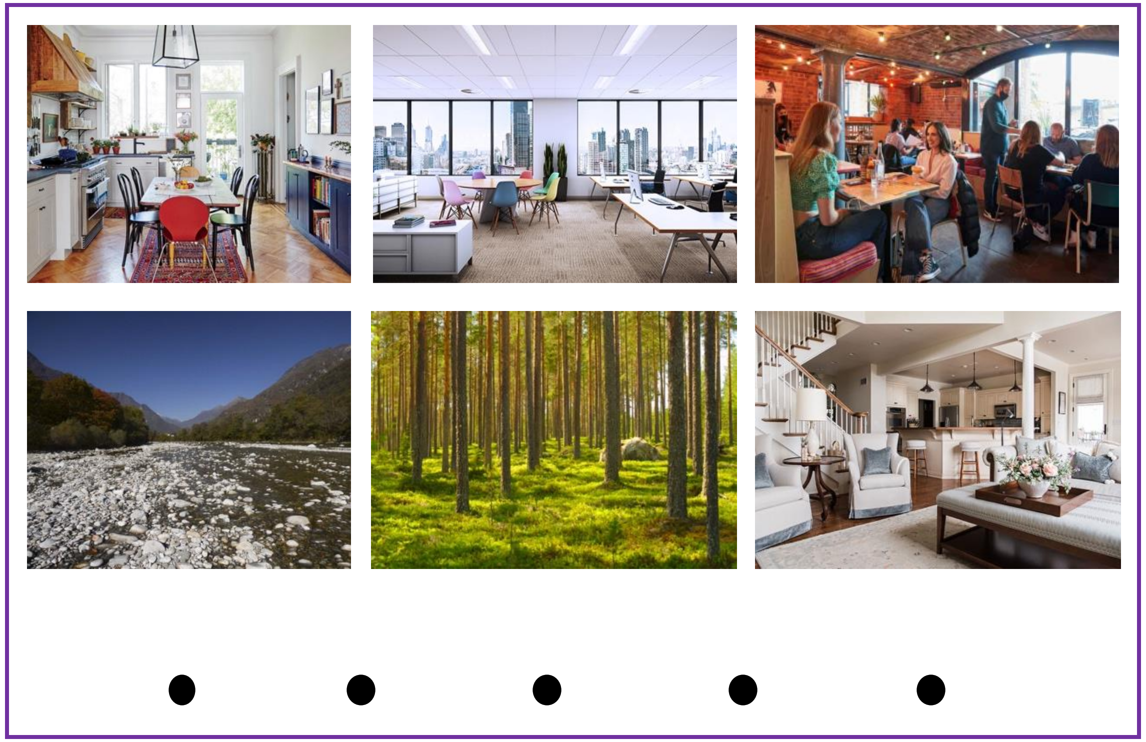
Robots that can do thousands of tasks in thousands of environments





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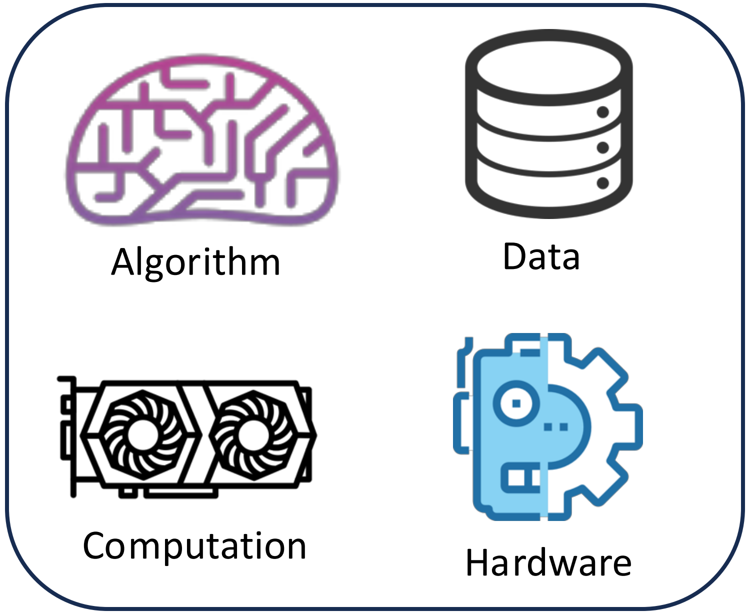

How far are we?

Embodied AGI in the real physical world

- We've made (a lot of) progress in terms of **domain-specific embodied intelligence**, either learning-based or not
- However, we are still very far from general-purpose embodied intelligence!

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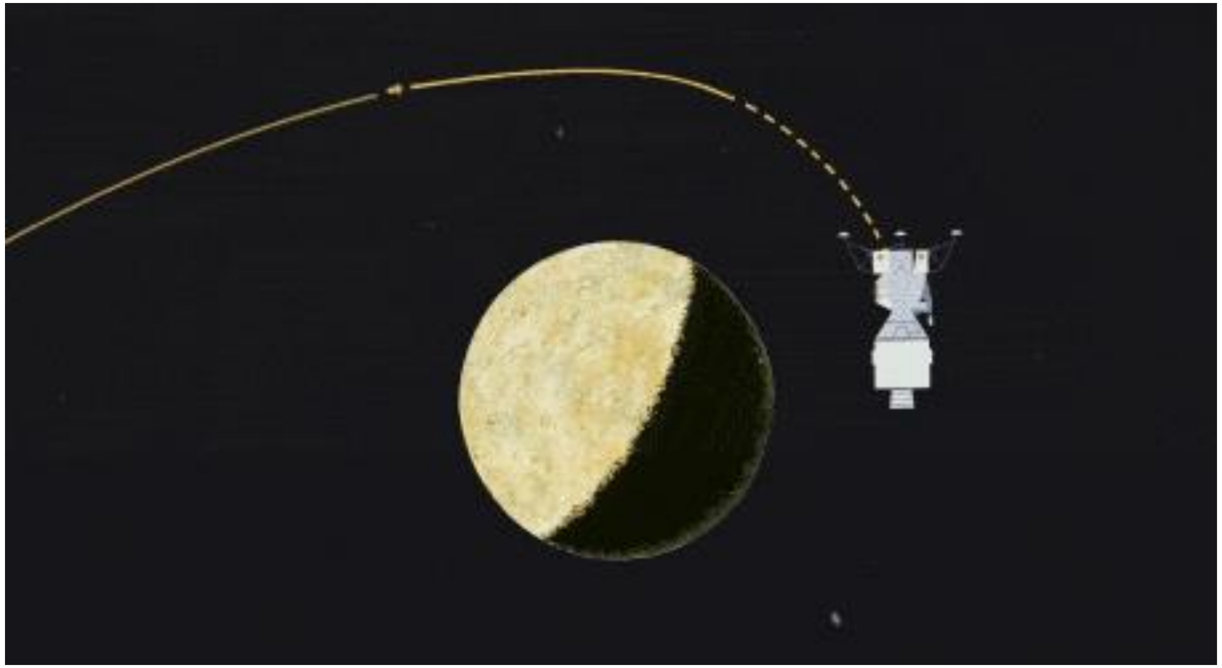
Where are We Today: Non-learning Method

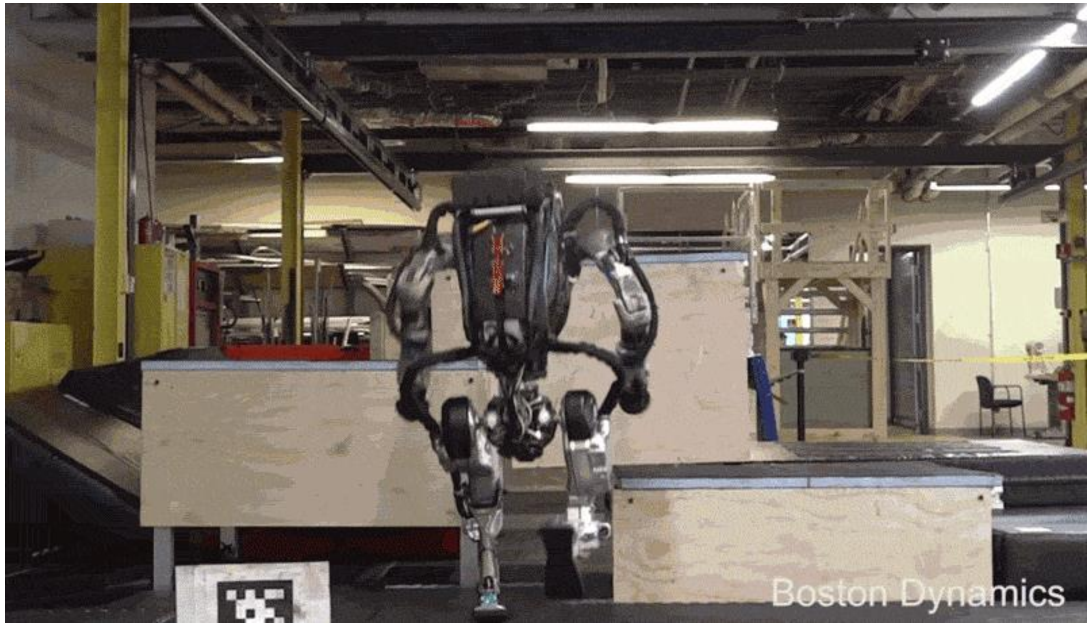
- Before seeing today's progress in robot learning, let's first see some results w/o learning:

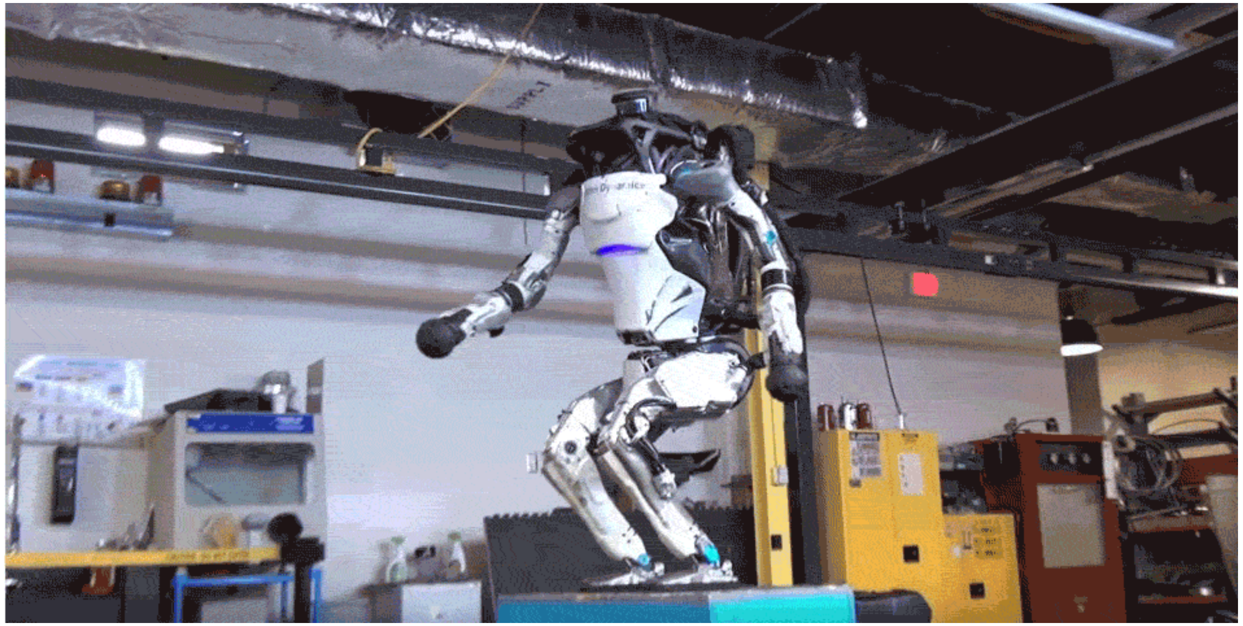


Apollo 11 trajectory optimization and control: optimal control + robust control

Where are We Today: Non-learning Method

- W/o learning:

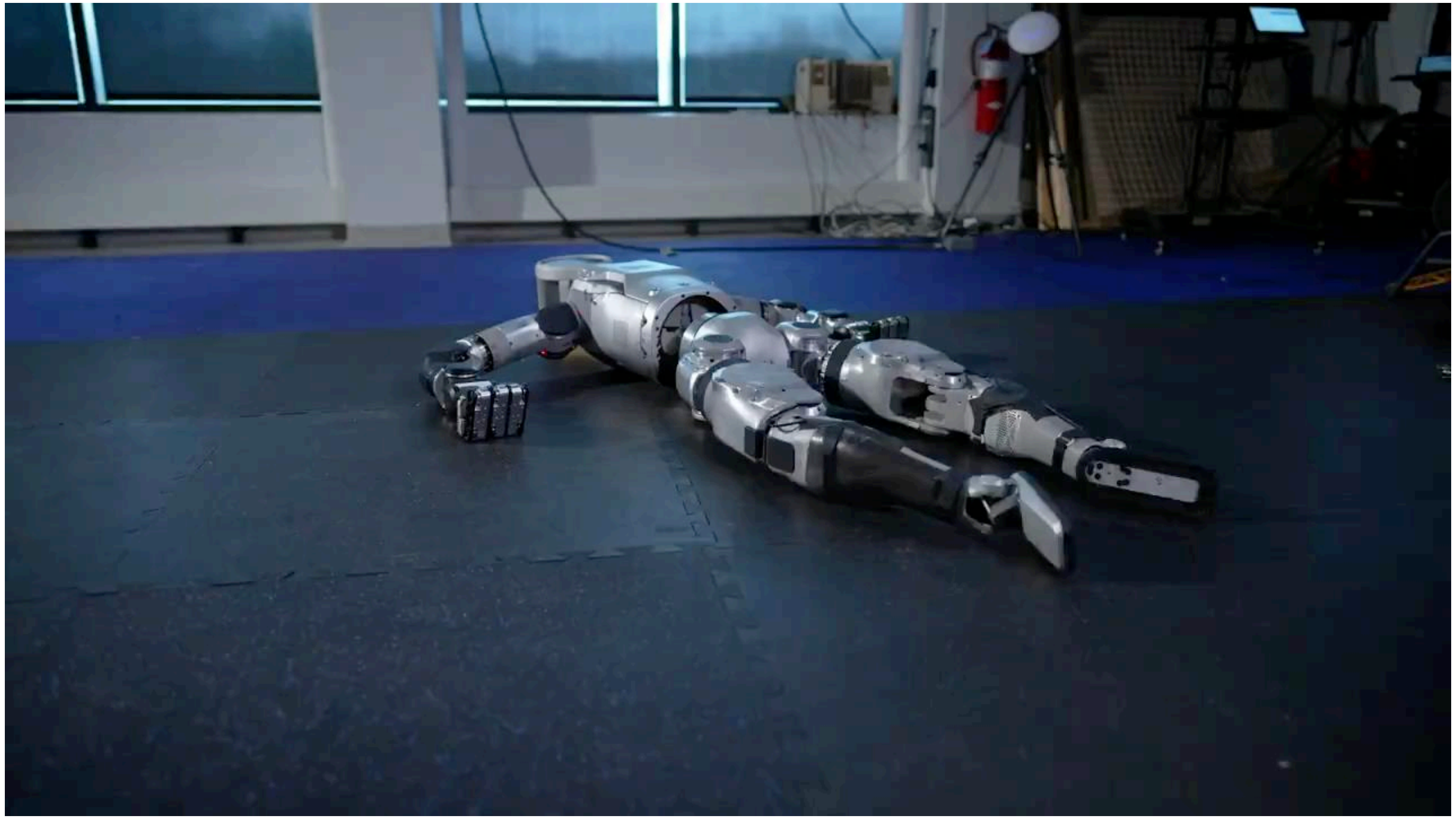




Boston Dynamics Atlas humanoid planning and control: trajectory optimization + MPC

Where are We Today: Non-learning Method

- The legendary Atlas just retired on April 16, 2024



New Atlas (100% electric motors)

Where are We Today: Non-learning Method

- W/o learning:

Model Predictive Control for Aggressive Driving Over Uneven Terrain

Tyler Han*, Alex Liu, Anqi Li, Alexander Spitzer, Guanya Shi, and Byron Boots

Offroad autonomy: sampling-based MPC

Where are We Today: Non-learning Method

- W/o learning:

High-speed robot hand (Ishikawa Komuro Lab) in 2009

Where are We Today: Non-learning Method

- W/o learning:

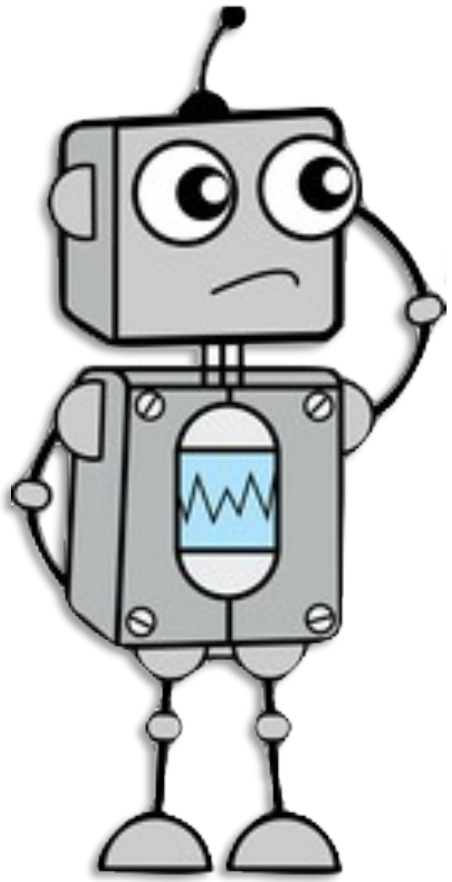
Tailsitter aerobatics (MIT LIDS): trajectory generation + nonlinear control

Why Do We Need Robot Learning?

- Data increases the “upper bound” of algorithm, computation, and hardware
  - Modeling the world could be hard
  - Environments/tasks might change
  - The policy structure might be limited
  - The optimization solver might be suboptimal
  - Assumptions might not hold

Why Do We Need Robot Learning?

❑ Non-learning methods often separately design each module



- Cognition
- Reasoning
- Planning
- Motion Skills
- Control
- Actuation

Semantic Intelligence

Physical/Motion Intelligence

Mechanical/Hardware Intelligence

Inspired by Jonathan Hurst from Agility Robotics

Why Do We Need Robot Learning?

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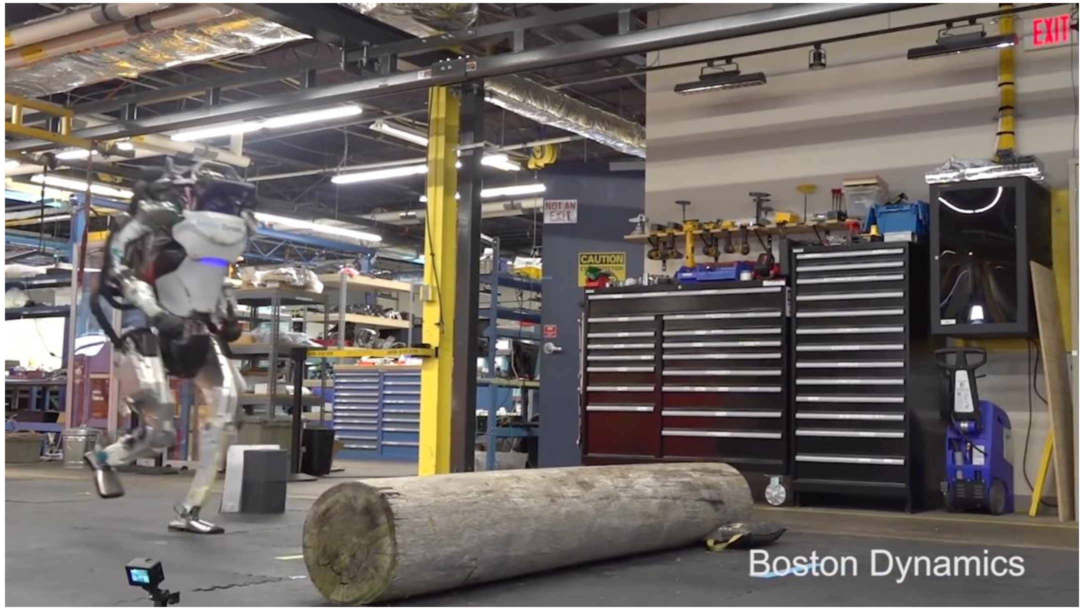
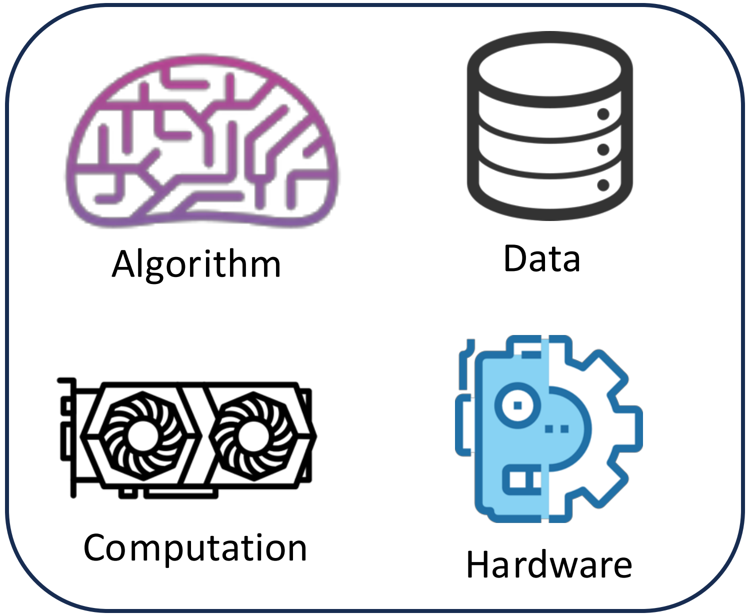


Next Time: A Glimpse of Today's Robot Learning

- In this class, we will learn how these methods work and more!
  - Model-free reinforcement learning (RL)
  - Sim2Real and Real2Sim2Real
  - Online adaptation
  - Imitation learning from demonstrations
  - Model-based RL
  - Offline RL



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Embodied AGI in the real physical world

Thank You!